

PEFT-Conditioned Data Selection for Cross-PEFT Data-Efficient Fine-Tuning

Anonymous submission

Abstract

Parameter-efficient fine-tuning (PEFT) reduces the number of trainable parameters, but efficient adaptation also depends on which examples drive training. Existing data selectors usually estimate one task-conditioned value per example and reuse the resulting subset across PEFT methods. This paper studies a structural limitation of that practice: PEFT methods expose different trainable subspaces, so the same example can have different utility under LoRA, adapters, and (IA)³.

We introduce PCU-Select, a PEFT-conditioned data selector that predicts sample utility from three conditions: the candidate example, a target-task sketch, and a structured PEFT configuration. PCU-Select aligns sample and task gradients on 24 intervention sites, reweights those sites by the target PEFT, corrects the proxy with short-horizon high-fidelity labels, and selects a coverage-aware subset with uncertainty penalties. Across four tasks, five seen PEFT configurations, and three target-training seeds, PCU-Select improves the 10% budget average over random selection by 3.64 points and over RDS+ by 0.98 points, while matching LESS on average. Its offline cost amortizes after about 1.6 target PEFT configurations relative to per-PEFT influence selection. Ablations confirm that PEFT encoding, task sketches, multi-fidelity supervision, and coverage-aware quotas all contribute to performance.

Introduction

Adapting large language models under limited compute requires efficiency on both the model side and the data side. Parameter-efficient fine-tuning (PEFT) freezes the backbone and trains a small set of task-specific parameters, reducing optimizer memory and deployment storage. Data-efficient fine-tuning selects a compact subset from a larger instruction pool, reducing the number of examples and training steps. These two tools are usually applied independently: PEFT controls which parameters move, while data selection controls which examples drive the update.

Current data selectors rarely model this interaction. Quality filters, diversity heuristics, proxy-model scores, and gradient-similarity methods often assign each example a single value for a target task and then reuse the selected

subset across adaptation methods. That assumption is convenient, but PEFT methods change the trainable subspace itself. LoRA injects low-rank additive updates into selected projections (Hu et al. 2022); adapters insert residual bottlenecks (Houlsby et al. 2019); and (IA)³ learns multiplicative activation scalings (Liu et al. 2022). An example whose gradient is aligned with a task inside one subspace need not be aligned inside another.

Our motivation study shows that PEFT-specific utility is not a small perturbation of a universal score. We estimate short-horizon utility for the same candidate pool under LoRA, adapter, and (IA)³ variants, using two task sketches and independent anchor/seed replicates. Figure 1 shows that within-PEFT agreement is much higher than cross-PEFT agreement. Same-family capacity changes retain moderate structure, but placement changes and cross-family changes sharply reduce rank agreement and Top-5% overlap. A transfer matrix gives the downstream consequence: selecting data for the same PEFT used in training beats mismatched selection by 4.98 points on average across target PEFTs.

This paper treats data value as a PEFT-conditioned quantity. We ask: given a candidate example x , a target task t , and a target PEFT configuration p , can we predict whether x will help when the trainable parameters are exactly those exposed by p ? This framing differs from PEFT-agnostic selection, which asks whether x is generally good for the task. It also differs from recomputing an influence selector for every PEFT, because the expensive supervision should be reused across many target configurations.

PCU-Select answers this question with an intervention-site utility layer. It maps samples, task sketches, and PEFT configurations into a shared set of 24 module-output sites in the backbone. Sample and task signatures describe gradient alignment at each site. The PEFT code describes which sites a method touches, how it acts there, and how much trainable capacity it has. PCU-Select uses a low-fidelity site-weighted gradient proxy for broad coverage, corrects it with short-horizon high-fidelity labels, and trains a conditional scorer that predicts both utility and uncertainty. At deployment, the scorer ranks the pool for a target (p, t) and a quota allocator balances utility with semantic coverage.

We make three contributions.

- **Problem and evidence.** We formulate cross-PEFT data-efficient fine-tuning and show that data-value rankings

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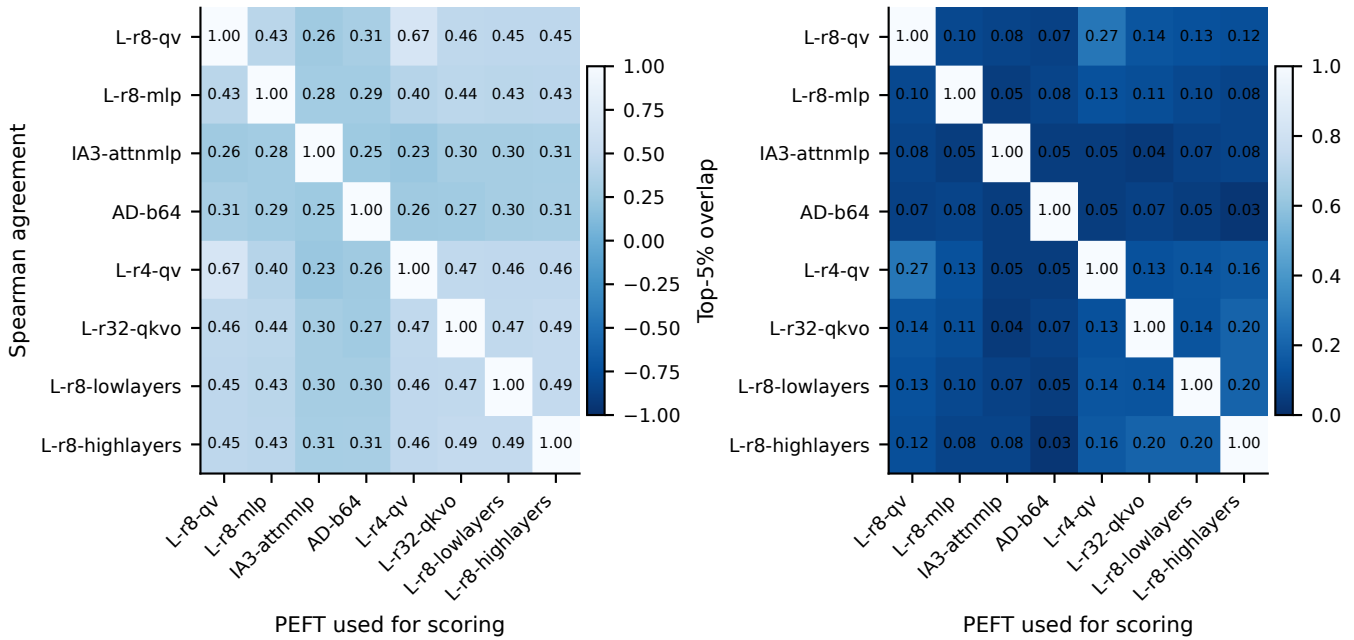


Figure 1: Short-horizon data-value estimates disagree across PEFT configurations. The same candidate examples and task sketches are scored under each target PEFT. Same-PEFT replicate agreement is high ($\rho = 0.79$, Top-5% overlap = 0.31), while cross-PEFT agreement falls with structural distance: cross-family pairs average $\rho = 0.28$ and Top-5% overlap = 0.06.

differ across PEFT configurations, with cross-family Top-5% overlap near 0.06 and a 4.98-point downstream mismatch penalty.

- **Method.** We introduce PCU-Select, a reusable PEFT-conditioned selector built from intervention-site gradient signatures, structured PEFT encodings, task sketches, multi-fidelity utility labels, uncertainty-aware scoring, and adaptive cluster quotas.
- **Evaluation.** Across GSM8K, HumanEval, MMLU, and TyDiQA, PCU-Select improves 10%-budget downstream performance over random by 3.64 points and RDS+ by 0.98 points. It matches LESS on average and amortizes its offline cost after about 1.6 target PEFTs.

Related Work

Instruction-data selection. Instruction tuning benefits from small but carefully selected data. LIMA shows that 1,000 curated examples can produce strong instruction-following behavior in a large model (Zhou et al. 2023). AlpGasus filters low-quality Alpaca examples with LLM judgments (Chen et al. 2024); InsTag measures semantic diversity and complexity through instruction tags (Lu et al. 2023); and DEITA combines quality, complexity, and diversity criteria for automatic data selection (Liu et al. 2024). Other selectors emphasize explicit diversity through quality-diversity trade-offs, log-determinant objectives, or partition-aware sampling (Bukharin et al. 2023; Wang et al. 2024; Rao et al. 2024). These methods improve data efficiency, but they mostly treat value as a property of the example and target task rather than of the PEFT subspace used for adaptation.

Proxy-model and influence selection. Model-assisted selectors score examples with external judges, weak models, or training signals. Cherry LLM uses instruction-following difficulty (Li et al. 2023); Superfiltering studies when weak models can rank data for stronger models (Li et al. 2024); and SmallToLarge transfers trajectory information from small models to larger targets (Yang et al. 2024). Influence-style methods estimate how training examples affect validation behavior. Classical influence functions use second-order approximations (Koh and Liang 2017), TracIn accumulates gradient similarities along checkpoints (Pruthi et al. 2020), and LESS builds a low-dimensional gradient datastore for instruction selection (Xia et al. 2024). DataInf specializes influence approximation for LoRA-tuned generative models (Kwon et al. 2024). PCU-Select uses gradient alignment as one signal, but it conditions the signal on the target PEFT configuration and learns a reusable scorer instead of rebuilding a per-PEFT selector.

Parameter-efficient fine-tuning. PEFT methods reduce adaptation cost by updating a small module or parameter subset. Adapters insert bottleneck modules into Transformer layers (Houlsby et al. 2019; Pfeiffer et al. 2021). Prefix and prompt tuning optimize continuous prompts or prefix states (Li and Liang 2021; Lester, Al-Rfou, and Constant 2021). LoRA learns low-rank additive updates (Hu et al. 2022), QLoRA combines quantization with LoRA for memory efficiency (Dettmers et al. 2023), BitFit tunes only bias terms (Ben Zaken, Goldberg, and Ravfogel 2022), and (IA)³ rescales attention and feed-forward activations (Liu et al. 2022). PEFT work usually compares accuracy, parameter count, memory, or inference overhead under fixed data. This

paper asks a complementary question: when the candidate pool is fixed, do different PEFT methods want the same examples?

Joint data- and parameter-efficient adaptation. Data selection and PEFT address two sides of efficient adaptation, but most work optimizes one side while holding the other fixed. DataInf and LESS move toward the intersection by using gradients from LoRA-style training (Kwon et al. 2024; Xia et al. 2024), and geometry-aware targeted selection suggests that PEFT optimization has nontrivial subspace structure (Min et al. 2026). PCU-Select positions the target PEFT itself as an input to data selection. Unlike PEFT-agnostic filters, it changes the selected subset when the trainable subspace changes; unlike per-PEFT influence methods, it amortizes the expensive supervision across target configurations.

Problem Formulation

We formalize *cross-PEFT data-efficient fine-tuning*: selecting a compact training subset whose value depends on the target PEFT configuration and target task. Fix a backbone family and frozen base model M_0 . A candidate pool $\mathcal{D} = \{x_i\}_{i=1}^N$ contains instruction-response examples. A task t provides a small validation sketch V_t drawn from train or development data, never from the test set. A PEFT configuration $p \in \mathcal{P}$ specifies the family, trainable locations, capacity, and training recipe. Given budget B , the ideal subset solves

$$\mathcal{S}^* = \arg \max_{\mathcal{S} \subseteq \mathcal{D}, |\mathcal{S}| \leq B} \text{Perf}(\text{Tune}(M_0, p, \mathcal{S}), V_t^{\text{test}}). \quad (1)$$

Directly optimizing Eq. (1) is infeasible because it requires training and evaluating many subsets for every target (p, t) pair.

PCU-Select replaces the subset oracle with a conditional per-example utility,

$$s_\phi(x, p, t) \approx u(x, p, t), \quad (2)$$

where u measures how much x helps task t when the trainable parameters are those of p . Conditioning on p is the core departure from PEFT-agnostic selection. A sample’s gradient can be useful in the attention projections trained by LoRA, weak in the residual bottlenecks trained by adapters, and different again under multiplicative (IA)³ gates. The selected subset should therefore change when the trainable subspace changes.

The formulation rests on four requirements. **PEFT dependence**: the scorer must take p as an explicit input. **Task relativity**: utility must be defined relative to V_t so that the model does not learn a generic quality score. **Structural encodability**: PEFT methods must be represented by their sites, operator type, capacity, and recipe, not just by a family name. **Amortizable cost**: expensive labels and scorer training are paid offline, while applying to a new (p, t) pair uses cached features and forward scoring.

Method: PCU-Select

PCU-Select learns the surrogate in Eq. (2) and uses it to select a budgeted subset for a target PEFT and task. Figure 2 summarizes the pipeline. The offline stage extracts reusable

sample and task signatures, builds low- and high-fidelity utility labels, and trains the scorer. The online stage encodes a target (p^*, t^*) , checks whether p^* is outside the training configuration distribution, scores the pool, and allocates the budget across semantic clusters.

Intervention Sites

PCU-Select compares PEFT methods through module-output sites rather than raw parameter tensors. For a decoder-only Transformer, it selects eight evenly spaced layers and hooks three outputs in each layer: attention output, MLP output, and post-residual block output. The resulting set Ω has 24 sites. Module outputs are stable across backbones with fused or split projection implementations, and they keep the cache size manageable.

Each PEFT configuration maps to a site mask and an operator type. Attention LoRA and (IA)³-attention touch attention-output sites; MLP LoRA and (IA)³-FFN touch MLP-output sites; adapters touch block-residual sites; prefix tuning touches attention-output sites; and BitFit is represented as a bias shift. PCU-Select assigns each site a normalized PEFT weight

$$\begin{aligned} \alpha_p^\omega &= \text{mask}(p, \omega) \rho_{\text{op}} c(p, \omega), \\ c(p, \omega) &= \tanh(\eta \log(1 + \text{cap}(p, \omega)/d_{\text{model}})), \\ \tilde{\alpha}_p^\omega &= \alpha_p^\omega / \max(\epsilon, \sum_{\omega'} \alpha_p^{\omega'}). \end{aligned} \quad (3)$$

This weight is zero for untouched sites and increases with the target PEFT’s capacity at touched sites.

Representations and Utility Labels

A sample representation $z_x = [e_x; d_x; a_x]$ concatenates a 768-dimensional semantic embedding, 16 difficulty and format statistics, and a 64-dimensional activation signature. A task sketch V_t is encoded by pooling the same sample representation over 32 sketch examples and by averaging their site gradient signatures. A PEFT code $z_p = [m_p; c_p; r_p]$ contains a 24×4 site/operator mask, a 16-dimensional capacity vector, and a 16-dimensional training-recipe vector.

Low-fidelity utility aligns each sample with the target sketch on the sites trained by the target PEFT:

$$u^{\text{lo}}(x, p, t) = \sum_{\omega \in \Omega} \tilde{\alpha}_p^\omega \cos(g_x^\omega, g_t^\omega). \quad (4)$$

Here g_x^ω is a unit-normalized random projection of the response-token loss gradient at site ω , and g_t^ω is the sketch mean. The proxy is cheap after sample signatures are cached, so it labels many (x, p, t) triples.

High-fidelity utility corrects the proxy near the selection boundary. For a sampled triple, PCU-Select performs short PEFT updates from a base anchor and a warm anchor, then measures sketch-loss reduction:

$$\Delta(x, p, t, a, h) = L_{V_t}(\theta_a) - L_{V_t}(\text{Adapt}_p^h(x; \theta_a)). \quad (5)$$

Rank-normalized reductions from horizons $h \in \{1, 4\}$ and both anchors define u^{hi} . The high-fidelity budget is 10K triples, split across coverage sampling, uncertainty sampling, and boundary sampling.

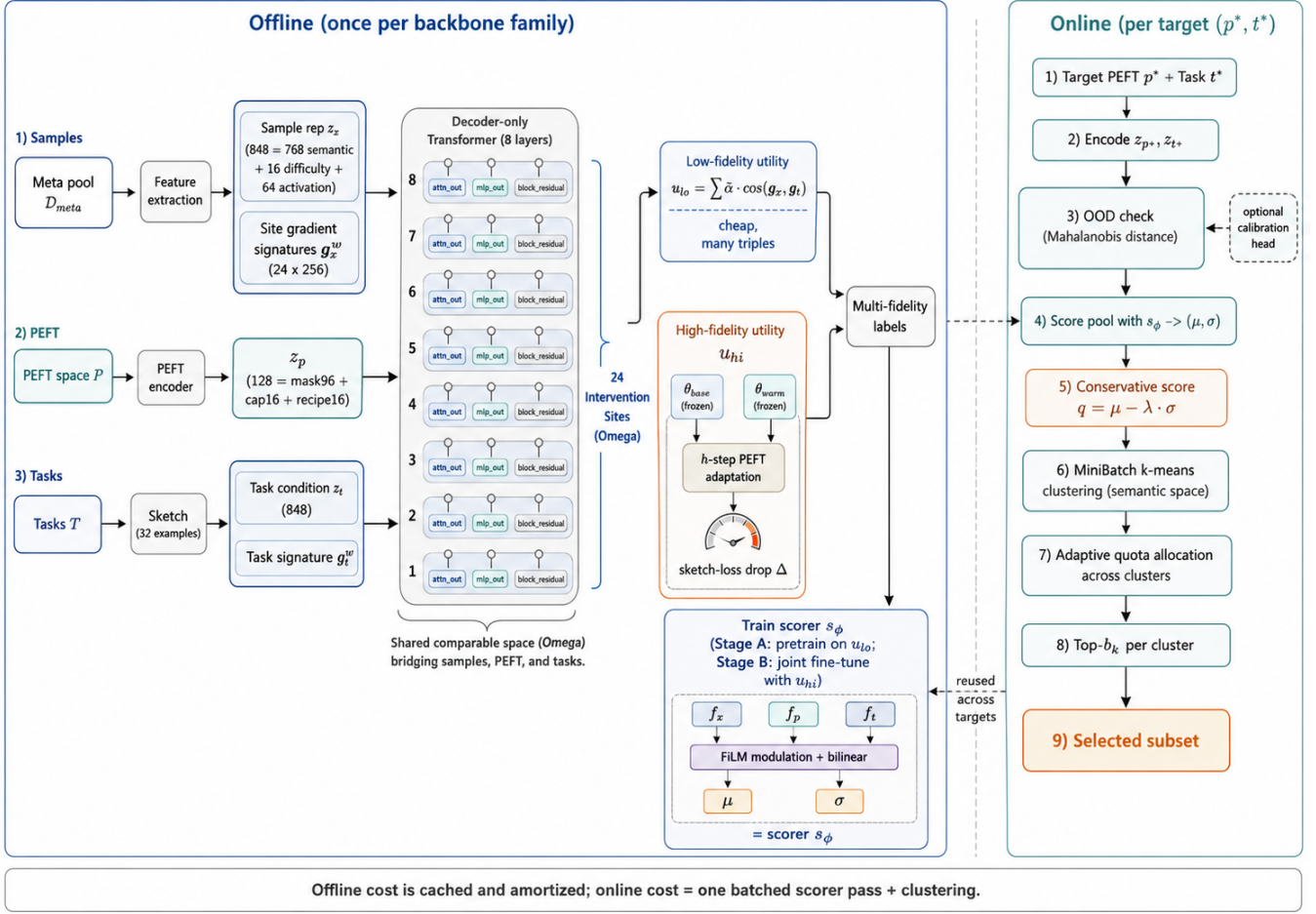


Figure 2: PCU-Select separates reusable offline supervision from cheap online selection. Intervention sites Ω provide the shared coordinate system that links sample gradients, task sketches, and PEFT configurations.

Scorer and Selection

The scorer predicts $(\hat{\mu}, \hat{\sigma}) = s_\phi(z_x, z_p, z_t)$. Three encoders map z_x , z_p , and z_t into hidden vectors. The PEFT and task codes modulate the sample code through FiLM, while a bilinear term models sample-PEFT interaction. Training combines high-fidelity ranking and regression losses, low-fidelity proxy distillation, and a heteroscedastic uncertainty loss. Stage A pretrains on the proxy for broad conditional structure; stage B enables all losses and selects checkpoints on held-out PEFT/task pairs.

At application time, PCU-Select ranks examples by a conservative score

$$q(x) = \hat{\mu}(x, p^*, t^*) - \lambda \hat{\sigma}(x, p^*, t^*), \quad \lambda = 0.2. \quad (6)$$

Pure top- k ranking can over-select near-duplicates, so PCU-Select clusters examples in semantic space and allocates budget across clusters:

$$b_k = \text{round} \left(B \frac{(v_k^+)^{\alpha} |C_k|^{1-\alpha}}{\sum_j (v_j^+)^{\alpha} |C_j|^{1-\alpha}} \right), \quad \alpha = 0.6. \quad (7)$$

The final subset takes the top- b_k examples by q inside each cluster. For unseen PEFTs, a Mahalanobis distance over z_p

flags out-of-distribution targets; optional calibration fits a lightweight residual head from a few hundred high-fidelity labels.

Experiments

The evaluation asks four questions: does PCU-Select improve selected-data performance, does it amortize cost across PEFTs, which components matter, and how does it behave on unseen configurations?

Setup

We evaluate on four main tasks: GSM8K, HumanEval, MMLU, and TyDiQA. Metrics use each task’s native scale: exact match, Pass@1, accuracy, or F1. The candidate pool contains 300K mixed instruction examples, and the main selection budget is 10%; budget sensitivity also uses 5% and 30%. Each method trains the same target backbone, PEFT recipe, number of steps, and evaluation protocol; only the selected subset changes. Reported downstream values average over three target-training seeds.

The seen PEFT set contains attention LoRA, MLP

Table 1: Representative PEFT configurations. Parameter counts come from the registry counter for the Llama2-7B backbone.

Config	Family	Params	% bb	Sites
L-r8-qv	lora	4.19M	0.062	8/24
L-r8-mlp	lora	4.19M	0.062	8/24
IA3-attnmlp	ia3	0.39M	0.006	16/24
AD-b64	adapter	16.78M	0.249	24/24
L-r4-qv	lora	2.10M	0.031	8/24
L-r32-qkvo	lora	33.55M	0.498	8/24

LoRA, (IA)³, and adapters; unseen tests vary rank, placement, learning rate, and family. Baselines include Random and Balanced-Random; heuristic selectors based on length, loss, perplexity, and IFD; representation selectors such as embedding-nearest-neighbor, diversity clustering, and RDS+; and gradient/influence selectors including a LESS-style per-PEFT gradient baseline. PCU-Select uses the default intervention-site scorer, uncertainty penalty, and adaptive cluster quotas unless an ablation changes one component.

Main Results

Table 2 shows that PCU-Select improves selected-data quality under the same 10% budget. Averaged across four tasks and five PEFT configurations, PCU-Select scores 42.81, compared with 39.17 for random selection, 41.83 for RDS+, 42.16 for the influence baseline, and 42.73 for LESS. The gain over random is 3.64 points and the gain over RDS+ is 0.98 points. PCU-Select matches LESS on average but does not dominate it uniformly: it wins 10 of 20 PEFT-task cells and loses on the MLP-only LoRA column by 0.73 points. This supports the intended claim of reusable competitive selection rather than strict per-cell superiority over a recomputed per-PEFT influence selector.

Ranking metrics follow the same pattern. At the 10% budget, PCU-Select reaches mean Spearman $\rho = 0.625$, NDCG@K = 0.703, and pairwise accuracy = 0.793 against high-fidelity held-out utility labels. LESS reaches $\rho = 0.612$, NDCG@K = 0.677, and pairwise accuracy = 0.765. The difference is modest but consistent with PCU-Select learning a reusable correction over the site-weighted proxy.

Amortized Cost

PCU-Select concentrates cost in cacheable offline artifacts: feature extraction, low-fidelity labels, high-fidelity labels, and scorer training. The measured offline cost is 11.2 GPU-hours, dominated by 4.5 GPU-hours of high-fidelity labeling. Online application costs 0.925 GPU-hours per target on average, including selection and fixed target training. Per-PEFT LESS costs 0.996 online GPU-hours plus 6.8 GPU-hours of target-specific gradient recomputation. Figure 3 shows that PCU-Select breaks even with LESS after 1.63 target PEFTs. It does not beat RDS+ or random on raw cost; its advantage over those baselines is performance, not cheaper selection.

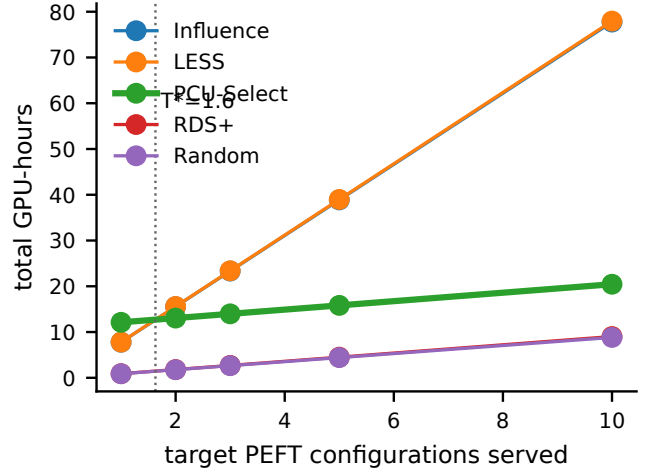


Figure 3: PCU-Select pays 11.2 GPU-hours offline, then applies cheaply to each target PEFT. It breaks even with per-PEFT LESS/influence recomputation after about 1.6 target configurations. Cheap PEFT-agnostic baselines remain cheaper in raw GPU-hours.

Analysis

PEFT Conditioning Changes the Selected Subset

Figure 4 tests whether the PEFT code changes behavior inside the same LoRA family. PCU-Select achieves 35.41 average performance across the E4 configuration scan, compared with 34.50 for LESS, 33.79 for RDS+, and 31.37 for random. The mismatch matrix shows that using the subset selected for the correct target configuration is better than reusing a subset selected for a different configuration by 2.42 points on average. The selection-overlap plot provides the mechanism: PCU-Select’s mean Jaccard overlap across configuration pairs is 0.427, while RDS+ remains 0.981 because it is PEFT-agnostic.

Ablations Identify the Useful Components

Table 3 shows that the largest drop comes from removing the PEFT code: performance falls by 2.45 points and NDCG falls from 0.702 to 0.526. Replacing the structured PEFT code with a family one-hot vector loses 1.22 points, which confirms that rank, placement, capacity, and recipe matter beyond the family name. Removing the task sketch loses 1.47 points, showing that PCU-Select is not only a PEFT-dependent quality filter. Multi-fidelity supervision also matters: low-fidelity-only training loses 1.52 points, while high-fidelity-only training loses 0.65 points because it lacks broad coverage. The adaptive cluster selector beats global top- k by 1.92 points and uniform cluster quotas by 0.91 points.

Unseen PEFTs Need Calibration

The Mahalanobis OOD check predicts when the offline scorer needs target-specific calibration. On L0 interpolation targets, zero-shot PCU-Select is within 0.34 points of LESS and calibration adds 0.61 points. On L1 extrapolated configurations, zero-shot PCU-Select trails LESS by 2.08 points and 500

Table 2: Main downstream performance at a 10% selection budget. Values average over four tasks and three target-training seeds; task-native metrics are scaled as percentages, so larger is better.

Method	AD-b64	IA3	L-r16-qkvo	L-r8-mlp	L-r8-qv	Avg.
Random	39.87	38.52	40.19	38.80	38.49	39.17
RDS+	42.86	40.75	42.55	41.87	41.13	41.83
Influence	42.75	41.15	42.90	42.10	41.93	42.16
LESS	43.23	41.82	43.68	42.72	42.22	42.73
PCU-Select	43.36	42.14	43.77	41.99	42.82	42.81

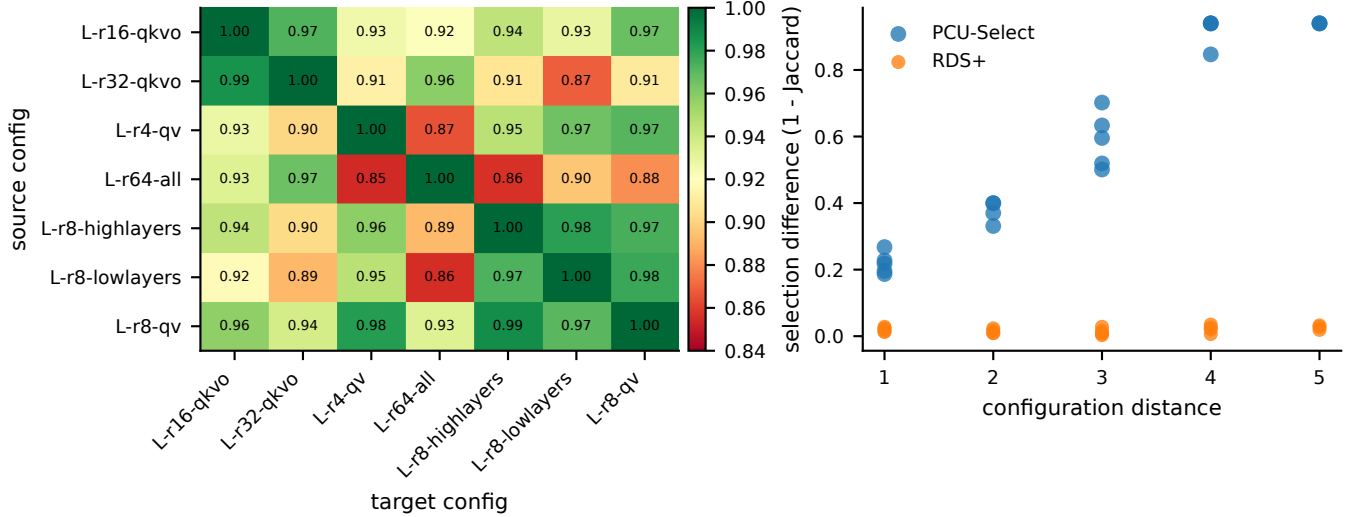


Figure 4: PCU-Select reacts to configuration changes. Left: subsets selected for one LoRA configuration degrade when used to train another, with an average 2.42-point diagonal-off-diagonal gap. Right: PCU-Select’s selection difference grows with configuration distance, while PEFT-agnostic RDS+ stays nearly flat.

Table 3: Ablations on GSM8K and HumanEval with two representative PEFTs at a 10% budget. Drop is relative to the full adaptive selector.

Variant	Metric	Drop	NDCG
Full PCU-Select	36.18	0.00	0.702
No PEFT code	33.73	2.45	0.526
Family one-hot only	34.96	1.22	0.602
No task sketch	34.71	1.47	0.597
No activation signature	35.32	0.87	0.629
Low-fidelity only	34.67	1.52	0.563
High-fidelity only	35.53	0.65	0.639
No uncertainty penalty	35.68	0.51	0.690
Global top- <i>k</i>	34.26	1.92	0.684
Uniform clusters	35.27	0.91	0.709

calibration labels recover 1.78 points. On L2 unseen families, zero-shot transfer fails more sharply: it trails LESS by 5.76 points, while 500 calibration labels recover 5.53 points. These results bound the main claim. PCU-Select is reliable for configurations near the training support and can be repaired with small calibration sets for farther targets; it is not a free zero-shot selector for arbitrary PEFT families.

Limitations

The current evidence covers one backbone family, four tasks, and a finite PEFT registry. The result bundle includes Qwen-style motivation assets, but the main E1–E5 quantitative claims are centered on the Llama2-7B registry. PCU-Select also uses short-horizon utility labels rather than full-training marginal contribution as supervision, so the scorer inherits the bias of that proxy. The cost claim is comparative. PCU-Select amortizes against per-PEFT influence recomputation, but cheap PEFT-agnostic selectors remain preferable when raw selection cost dominates and a small performance loss is acceptable.

Conclusion

Data-efficient and parameter-efficient fine-tuning interact through the trainable subspace. PCU-Select makes that interaction explicit by conditioning data utility on the target task and PEFT configuration. Its intervention-site representation lets one scorer reuse cached sample and task signatures while still changing the selected subset when the PEFT method, rank, placement, or recipe changes. The experiments show that this conditioning improves over PEFT-agnostic selection, matches per-PEFT LESS on average, and amortizes the expensive utility construction across multiple target con-

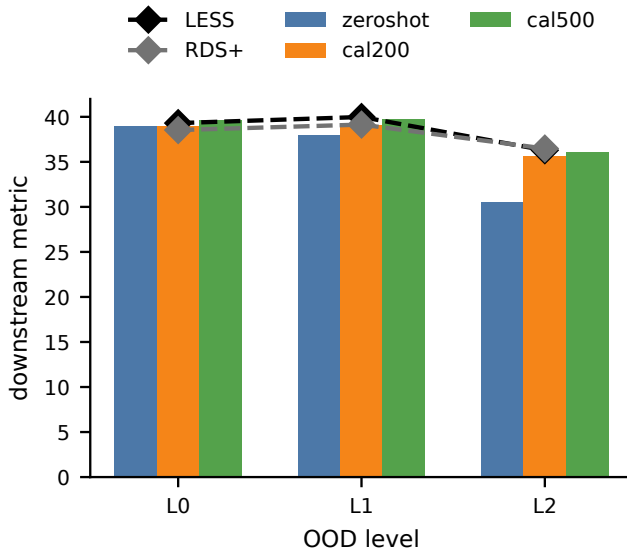


Figure 5: Zero-shot PCU-Select transfers to interpolated configurations, degrades on extrapolated and unseen-family PEFTs, and recovers with calibration labels. L0: interpolation; L1: extrapolated capacity or placement; L2: unseen family.

figurations. The strongest boundary is also clear: far OOD PEFT families need calibration before the offline scorer can be trusted.

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